

Exercise 3 SQL Case Statement

Question1

```
SELECT product_name,  
       Price,  
       CASE  
           WHEN price>1000 THEN 'Expensive'  
           WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'  
           WHEN price<100 THEN 'budget'  
       END AS price_category  
FROM products;
```

product_name	price	price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

Question2

```
SELECT customer_name,  
       amount,  
       CASE  
           WHEN amount>=1000 THEN 'High Value'  
           WHEN amount BETWEEN 500 AND 999 THEN 'Medium value'  
           WHEN amount<500 THEN 'Low Value'  
       END AS order_value_category  
FROM orders;
```

customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value

Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

Question3

```

SELECT emp_name,
       Department,
       Salary,
CASE
    WHEN department='IT' AND salary>8000 THEN 'Senior IT'
    WHEN department='HR' AND salary>55000 THEN 'Experienced IT'
    ELSE 'Staff'
END AS position_level
FROM employees;

```

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

Question4

```

SELECT student_name,
       Score,
CASE
    WHEN score>=90 THEN 'A'
    WHEN score BETWEEN 80 AND 89 THEN 'B'
    WHEN score BETWEEN 70 AND 79 THEN 'C'
    WHEN score BWTWEEN 60 AND 69 THEN 'D'

```

WHEN price<100 THEN 'F'

END AS grade

FROM students;

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

Question5

SELECT delivery_id,

delivery_time_minutes,

CASE

WHEN delivery_time_minutes<=30 THEN 'Fast'

WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'

WHEN delivery_time_minutes> 60 THEN 'LATE'

END AS performance,

FROM deliveries;

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

Question6

SELECT issue_type,

Priority,

CASE priority

WHEN 3 THEN 'High'

WHEN 2 THEN 'Medium'

WHEN 1 THEN 'Low'

END AS priority_label

FROM tickets,

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

Question7

SELECT student_id,

(days_present* 100 / today days) AS attendance_percentage,

CASE

WHEN (days_present* 100/ today days)>=90 THEN 'Excellent'

WHEN (days_present* 100/ today days) BETWEEN 75 AND 89 THEN 'Good'

WHEN (days_present* 100/ today days<75 THEN 'Needs Improvement'

END AS attendance_status

FROM attendance,

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

Question8

```
SELECT product id,  
        stock_qty,  
  
CASE  
  
    WHEN stock_qty=0 THEN 'Stock'  
  
    WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'  
  
    WHEN stock_qty > 5 'In Stock'  
  
END AS stock_status  
  
FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Question9

```
SELECT subject,  
        enrolled_students  
  
CASE  
  
    WHEN enrolled_students >= 25 THEN 'Large',  
  
    WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'  
  
    WHEN enrolled_students < 10 THEN 'Small'  
  
END AS class_size_category  
  
FROM classes;
```

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large

Question10

```
SELECT payment_id,  
       payment_method,  
CASE  
WHEN payment_method = 'Cash' AND amount >= 200 THEN 'Eligible for Discount'  
WHEN payment_method<> 'Cash' THEN 'Not Eligible'  
END AS discount_eligibility  
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount